



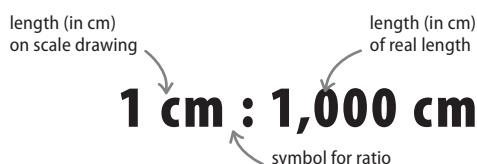
# Scale drawings

A SCALE DRAWING SHOWS AN OBJECT ACCURATELY AT A PRACTICAL SIZE BY REDUCING OR ENLARGING IT.

Scale drawings can be scaled down, such as a map, or scaled up, such as a diagram of a microchip.

## Choosing a scale

To make an accurate plan of a large object, such as a bridge, the object's measurements need to be scaled down. To do this, every measurement of the bridge is reduced by the same ratio. The first step in creating a scale drawing is to choose a scale—for example, 1 cm for each 10 m. The scale is then shown as a ratio, using the smallest common unit.



### SEE ALSO

◀ 56–59 Ratio and proportion

◀ 104–105 Enlargements

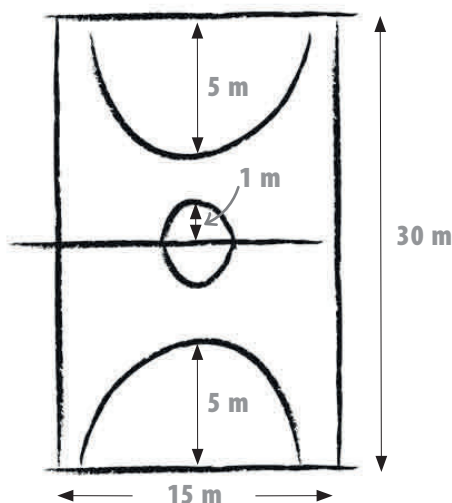
Circles 138–139 ▶

### Scale as a ratio

A scale of 1 cm to 10 m can be shown as a ratio by using centimeters as a common unit. There are 100 cm in a meter, so  $10 \times 100 \text{ cm} = 1,000 \text{ cm}$ .

## How to make a scale drawing

In this example, a basketball court needs to be drawn to scale. The court is 30 m long and 15 m wide. In its center is a circle with a radius of 1 m, and at either end a semicircle, each with a radius of 5 m. To make a scale drawing, first make a rough sketch, noting the real measurements. Next, work out a scale. Use the scale to convert the measurements, and create the final drawing using these.



**Draw a rough sketch** to act as a guide, marking on it the real measurements. Make a note of the longest length (30 m). Based on this and the space available for your drawing, work out a suitable scale.

Since 30 m (the longest length in the drawing) needs to fit into a space of less than 10 cm, a convenient scale is chosen:

measurement on drawing → **1 cm : 5 m** ← measurement on real court

By converting this to a ratio of 1 cm : 500 cm, it is now possible to work out the measurements that will be used in the drawing.

|                         | real measurements changed from meters to centimeters to make calculation easier | scale | length for drawing |
|-------------------------|---------------------------------------------------------------------------------|-------|--------------------|
| length of court         | $= 3,000 \text{ cm} \div 500$                                                   |       | <b>6 cm</b>        |
| width of court          | $= 1,500 \text{ cm} \div 500$                                                   |       | <b>3 cm</b>        |
| radius of center circle | $= 100 \text{ cm} \div 500$                                                     |       | <b>0.2 cm</b>      |
| radius of semicircle    | $= 500 \text{ cm} \div 500$                                                     |       | <b>1 cm</b>        |

**Choose a suitable scale** and convert it into a ratio by using the lowest common unit, centimeters. Next, convert the real measurements into the same units. Divide each measurement by the scale to find the measurements for the drawing.

